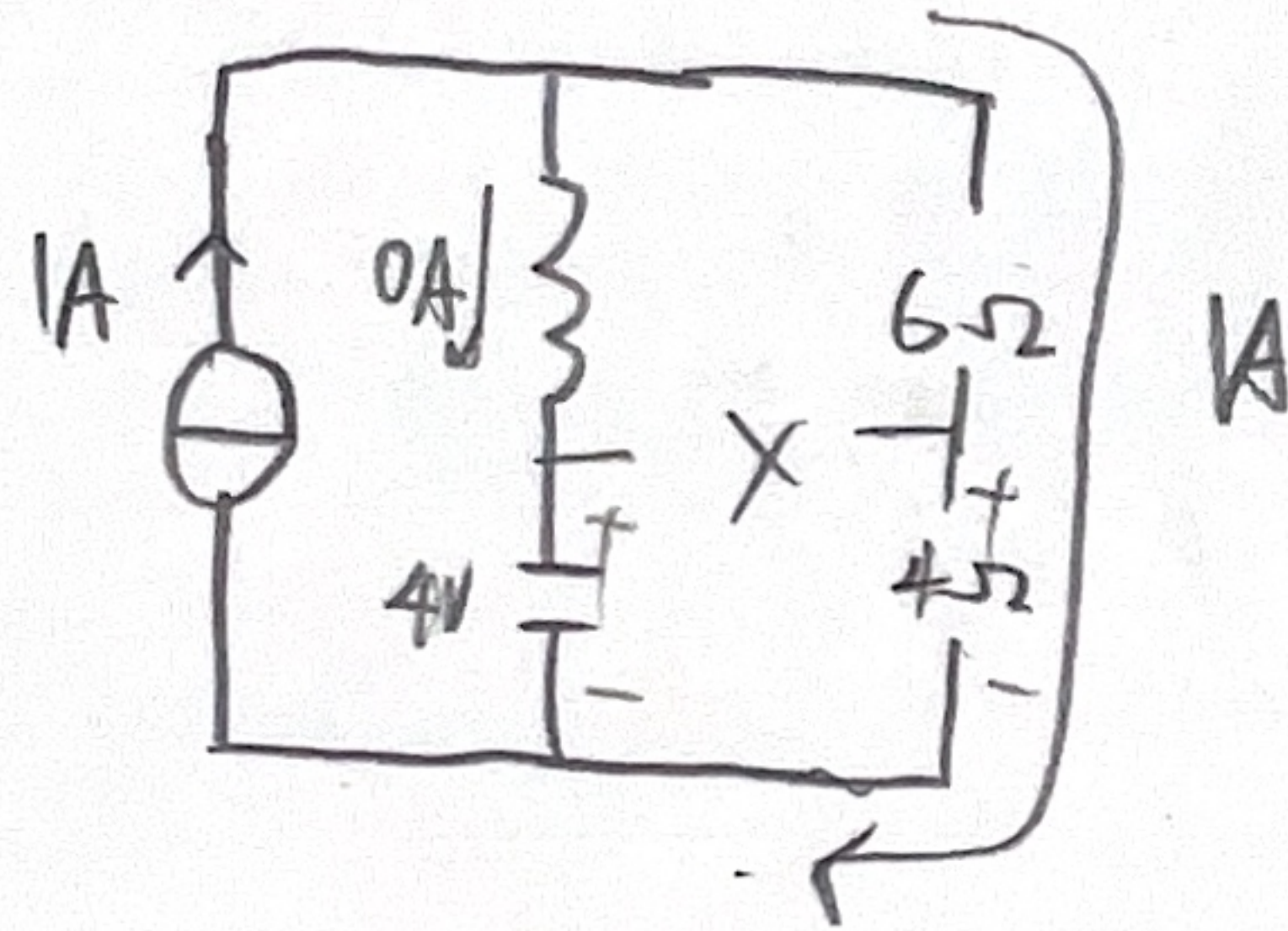
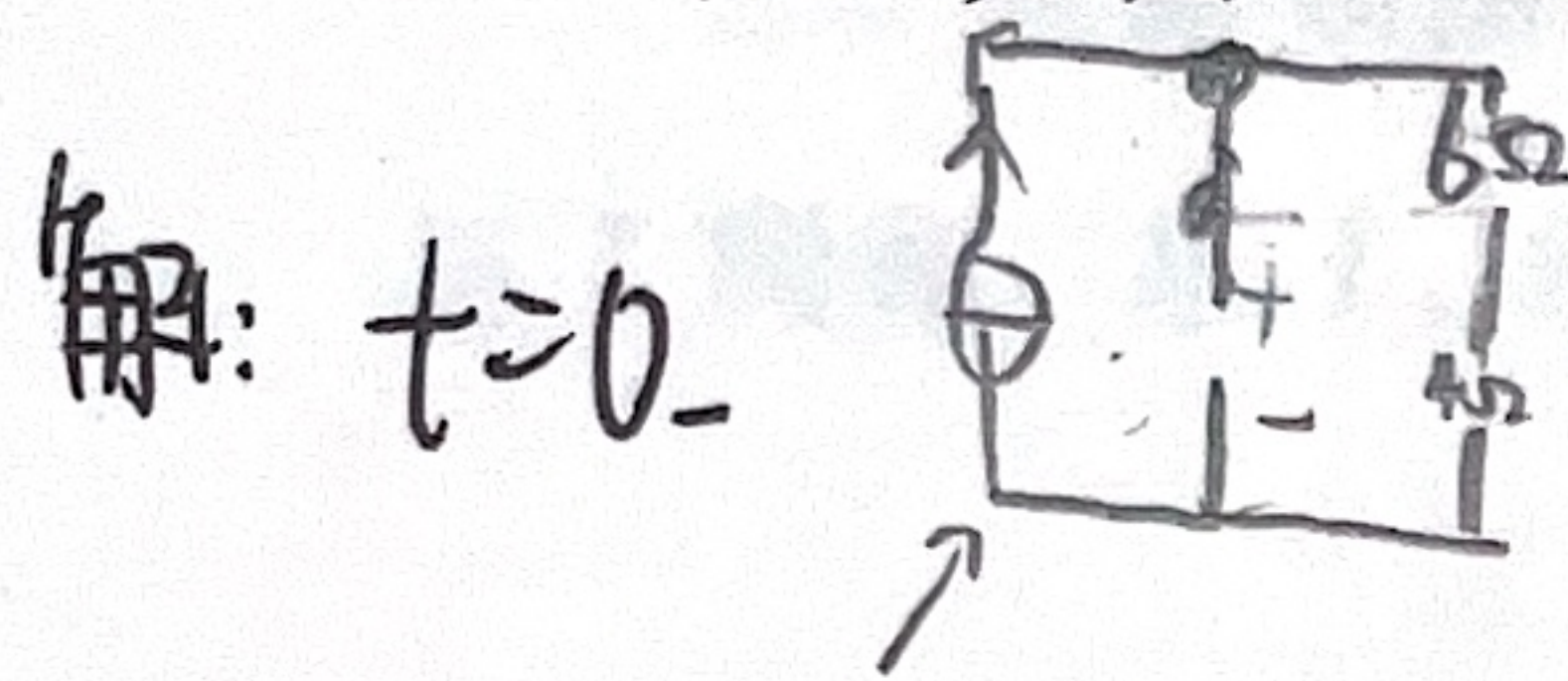
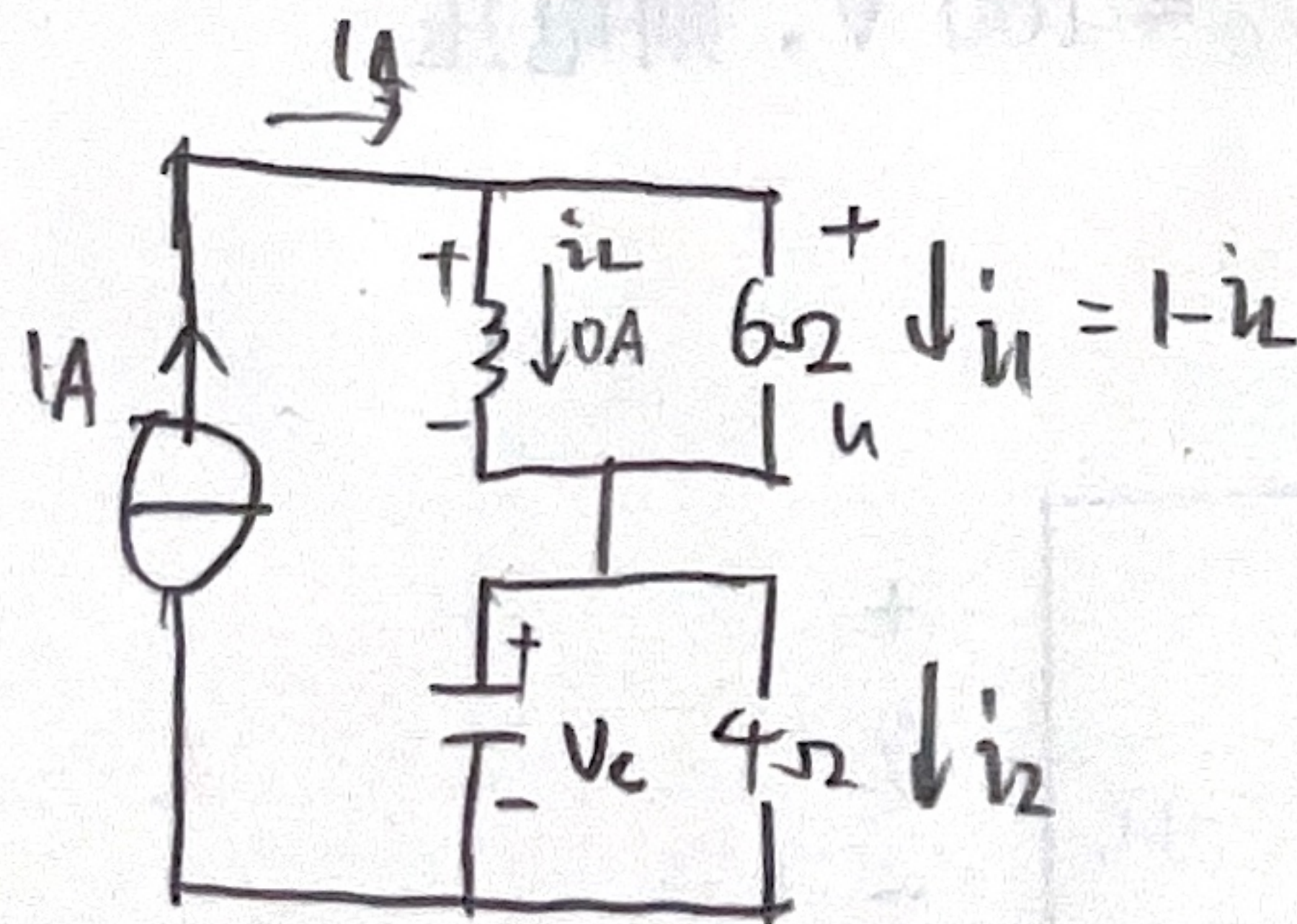


四、如图 3.4.5 所示电路已处于稳态， $t=0$ 时开关 S 闭合，求 $t \geq 0$ 时的电压 $u(t)$ 。

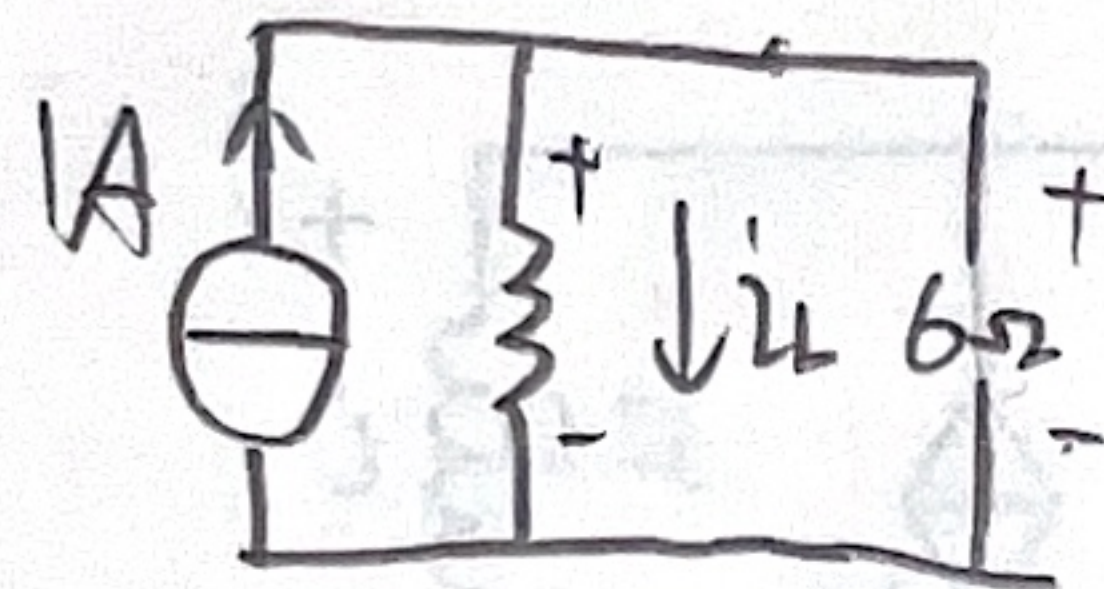


$t=0_+$

电路可等效为



由替代定理



$$i_L(0_+) = 0A$$

$$i_L(\infty) = 1A$$

$$\tau = \frac{L}{R_0} = \frac{2}{6} = \frac{1}{3}s$$

$$i(t) = 1 + (0 - 1)e^{-3t}$$

$$i(t) = (1 - e^{-3t})A$$

$$U_L(t) = 6[1 - 1 + e^{-3t}]A$$

$$U_L(t) = 6e^{-3t}A$$

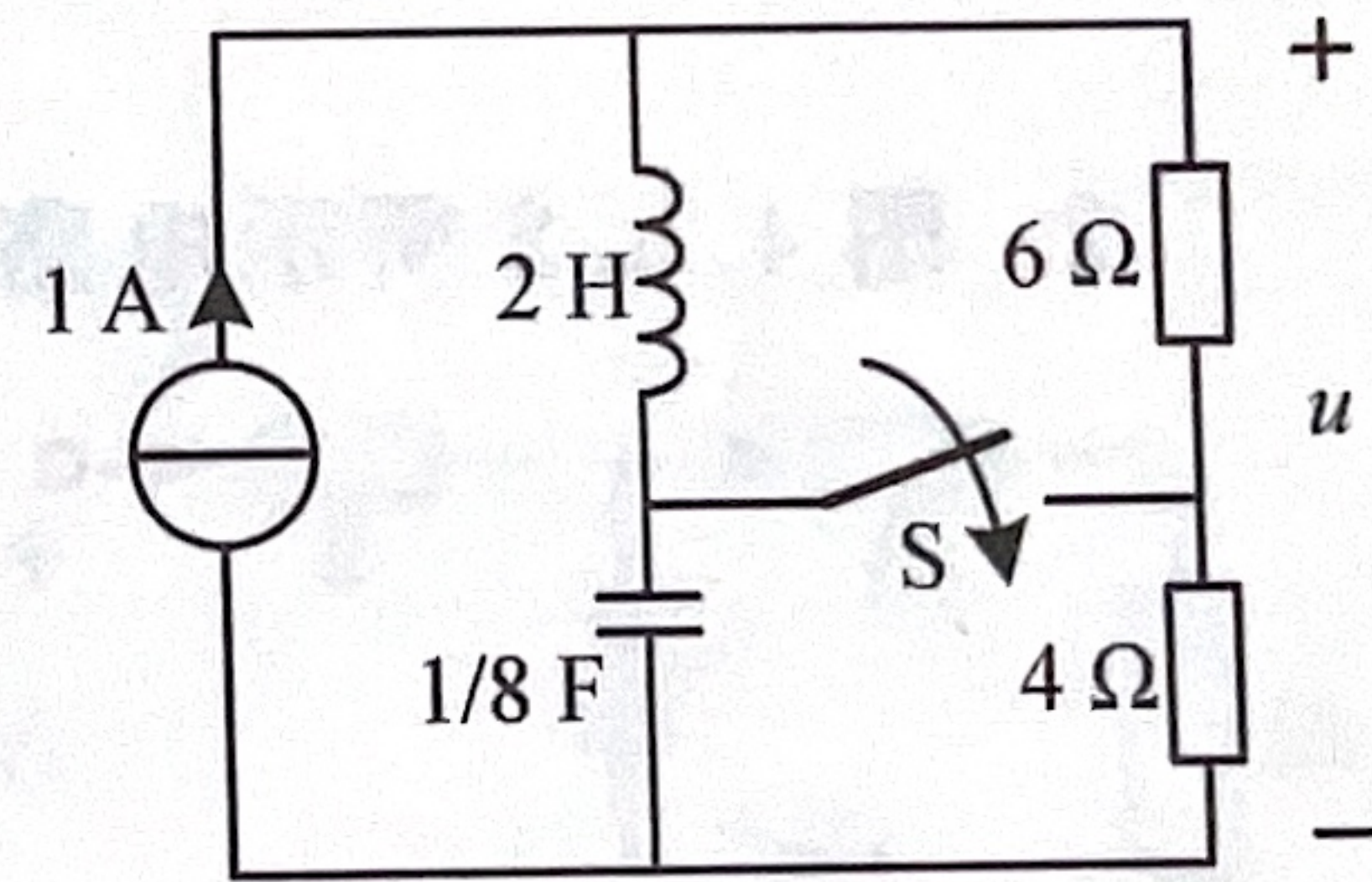
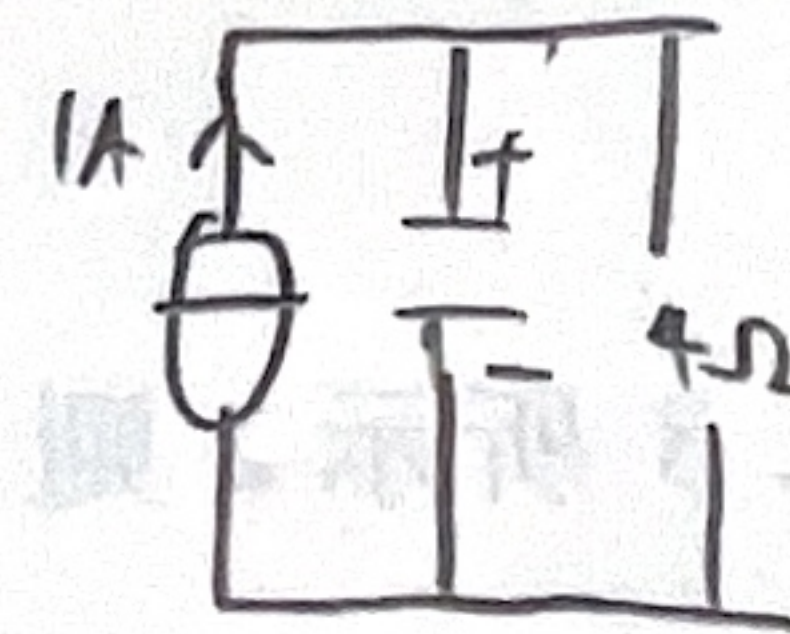


图 3.4.5



$$U_C(0_+) = 10V$$

$$U_C(\infty) = 4V$$

$$\tau = R_0 C = \frac{1}{2}s$$

$$U_C(t) = 10 + (4 - 10)e^{-2t}$$

$$U_C(t) = 4 + [10 - 4]e^{-2t} = 4 + 6e^{-2t}$$

$$U = U_L(t) + U_C(t)$$

$$U = (4 + 6e^{-3t} + 6e^{-2t})V$$

$t > 0$